

Hein Zarni Aung

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EDUCATION

Stony Brook University

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Full Stack Development, Software Security, Algorithms

Expected Graduation: May 2027

RESEARCH

Undergraduate Research Assistant — Ethos Lab, SBU

PI: Abisheka Pitumpe | Team size: 2

- Built a pipeline to monitor newly issued SSL/TLS certificates in real time and flag potentially fraudulent domains impersonating EZ-Pass tolling services, scraping keywords from the raw HTML, CSS, JS, and URLs of newly created sites.
- Used Selenium to automate certificate and domain scraping and Python to filter and triage flagged sites; auto-captured homepage screenshots of promising candidates to speed up manual review.

Sep 2025 – Dec 2025 (paused)

INTERNSHIPS

Software Engineering Intern — Clustr

Cross-platform React Native social/scheduling app | ~8-person dev team

- Contributed to Clustr, a cross-platform React Native social/scheduling app, delivering 50+ commits across backend and frontend features.
- Built server-side content and product data infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows, like/dislike interactions, and activity feed endpoints.
- Designed and implemented a Scheduler UI module featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, custom date/time picker integration, and a calendar-completion animation to improve user engagement.
- Implemented secure authentication features, including a password reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.
- Collaborated with an 8-person, multi-developer team via Git, regularly merging and resolving conflicts across feature branches to keep community and scheduling modules in sync with the shared codebase.

Jan 2026 – Sept 2026 (expected)

Software Engineering Intern — Prenora Dynamics LLC

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors to schedule and host lessons in one platform, and built a seed-data workflow to streamline debugging and local development.

Jun 2026 – Sept 2026 (expected)

CAMPUS INVOLVEMENT

Teaching Assistant - Data Structures & Algorithms, Dept. of Computer Science, SBU

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original PowerPoint-based review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

Aug 2025 – Present

International Peer Mentor — Office of Global Affairs, SBU

- Mentored 6 international students as part of the university's International Student Success peer mentoring program, through biweekly one-on-one meetings tailored to individual needs.

Aug 2025 – Present

PROJECTS

TuneAcademy — Hack Brooklyn 2026 (36-hour hackathon)

Winner: Best Prenora Dynamics Hack (Sponsor Challenge) | Team of 4

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Pivoted from a planned machine learning model to rule-based scoring algorithms under hackathon time and compute constraints, delivering reliable weakness detection within the 36-hour build window.
- Collaborated in a 4-person team; broader app included a React/Tailwind frontend, Python/FastAPI backend in Docker, Cloud Firestore, GCP hosting, and ElevenLabs-generated voice feedback reports.

Apr 2026

SKILLS

Languages: JavaScript, TypeScript, Python, Java, C

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium